

A
Project Report
On
Air Ticket Reservation System

Submitted By:

Your Name

SEAT NO:

Project Submitted To:



[NAAC Re-Accredited Grade: 'B' 2nd Cycle]

Affiliated To:

**Kavayitri Bahinabai Chaudhari North Maharashtra
University, Jalgaon Mah 425001**

For the Partial:

In the partial fulfillment of the requirement for the award of the degree

Bachelor of Computer Applications (3rd Year)

BCA 2025-26



**Dhanaji Nana Chaudhari Vidya Prabodhini Sanchalit
Shirish Madhukarrao Chaudhari College, Jalgaon.**

CERTIFICATE

This is to certify that **Your Name**, a student of **Bachelor of Computer Applications (BCA)** from **Dhanaji Nana Chaudhari Vidya Prabodhini Sanchalit Shirish Madhukarrao Chaudhari College, Jalgaon**, has successfully completed the project titled **Air Ticket Reservation System** under the guidance of **(Guide Name)** during the **academic year 2025–26**.

This project report has been submitted in partial fulfillment of the requirements for the award of the degree **Bachelor of Computer Applications (BCA)** under **Kavayitri Bahinabai Chaudhari North Maharashtra University, Jalgaon 425001**.

Project Guide

Guide Name

Principal

Principal / College name

Head of Department

HOD Name

Internal Examiner

External Examiner

DECLARATION

I hereby declare that the field project report titled **Air Ticket Reservation System** submitted by me to **Dhanaji Nana Chaudhari Vidya Prabodhini Sanchalit Shirish Madhukarrao Chaudhari College, Jalgaon**, in partial fulfillment of the requirements for the degree of **Bachelor of Computer Applications (BCA)**, is an original work carried out by me under the guidance of **(Guide Name)**.

I further declare that this work has not been submitted to any other university or institution for the award of any degree or diploma. All sources of information used in this project have been properly acknowledged.

Submitted By:

Your Name

SEAT NO:

BCA 2025–26

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to all those who have supported and guided me throughout the successful completion of my project titled **Air Ticket Reservation System**

I am especially thankful to my project guide, (Guide Name), for their invaluable guidance, continuous encouragement, and constant support during the development of this project. Their insightful suggestions, expert advice, and constructive feedback played a significant role in shaping this work and helped me complete it successfully.

I would also like to extend my heartfelt thanks to the faculty members of the department of computer science & applications for providing the necessary resources, knowledge, and academic support required for this project.

ABSTRACT

The Air Ticket Reservation System is a web-based application developed using PHP, MySQL, HTML, CSS, and JavaScript. The system provides a complete online platform for passengers to search for available flights, select seats, enter passenger details, and confirm bookings with payment. A dedicated administrator panel allows management of flights including adding new flights, updating fare and seat availability, handling delays and cancellations with automatic time recalculation, and viewing all passenger bookings.

The project demonstrates the practical use of a three-tier client-server architecture with session-based security, relational database design, server-side scripting, and front-end interactivity. The system replaces manual, paper-based ticketing processes and provides 24/7 online access to passengers and administrators alike.

INDEX

Sr. No.	Chapter Name	Page No.
1.	Introduction	7
2.	Requirement Analysis	10
3.	Proposed System	13
4.	Design and Architecture	15
5.	Development and Coding	29
6.	Testing and Implementation	33
7.	Conclusion	36
8.	Future Work	37
9.	References	38

CHAPTER 1

INTRODUCTION

The Air Ticket Reservation System is a web-based application developed using PHP, MySQL, HTML, CSS, and JavaScript. It allows passengers to search for available flights based on source city, destination city, and travel date, select seats, fill in passenger details, and complete the booking process online. The system also includes a comprehensive admin panel for managing flights, updating their status, and monitoring all bookings.

This project aims to replace manual, paper-based ticketing processes with a fast, reliable, and user-friendly digital platform. It follows a two-tier architecture separating the Admin and User roles, each secured by session-based authentication. The system is designed to be simple to use, requiring no technical knowledge from the end user.

In today's fast-paced world, online booking systems have become essential for the travel industry. This system serves as a complete solution for managing flight bookings with real-time seat availability, allowing passengers to book tickets from the comfort of their homes. The administrator has full control over the flight inventory, enabling efficient management of all flights and passenger data from a centralised dashboard.

1.1 SCOPE OF THE PROJECT

The objective of this application is to provide an online platform for passengers to search and book air tickets conveniently. It also provides the airline administrator with a centralised system to manage all flights, control seat inventory, update delays and cancellations, and view all booking records. The system covers the following scope:

- Online flight search by source, destination, and date.
- User registration, login, and OTP-based password recovery.
- Seat selection and passenger detail entry for each traveller.
- Payment summary and booking confirmation.
- Admin control over flights: add, view, edit, update delay and cancellation status.
- Booking history accessible per user based on email session.

1.2 EXISTING SYSTEM

The existing systems for small-scale airline reservation are largely manual or disconnected. They lack real-time updates for flight delays, do not provide online seat selection, and offer no self-service booking history for passengers. Administrators have to manage records manually, which leads to errors and data loss. Key problems include:

- No online registration or profile management for users.
- Manual record keeping leads to data loss and mismanagement.
- No real-time flight status updates for delays or cancellations.
- No centralised database for booking records and passenger information.
- Lack of internet-based access makes the system inaccessible remotely.

1.3 FEASIBILITY STUDY

A feasibility study is a high-level capsule version of the entire system analysis and design process. The study begins by classifying the problem definition. Feasibility is to determine if it is worth doing. There are 3 parts in the feasibility study:

1.3.1 Operational Feasibility

Operational feasibility is the measure of how well a proposed system solves the problems and takes advantage of the opportunities identified during scope definition. The system is operationally feasible as it replaces time-consuming manual booking with a digital, self-service platform. Both users and admins can interact with the system through a standard web browser without requiring specialised training. The operational feasibility assessment focuses on the degree to which the proposed development project fits in with the existing business environment and objectives with regard to development schedule, delivery date, and existing business processes.

1.3.2 Technical Feasibility

Technical feasibility involves questions such as whether the technology needed for the system exists and how difficult it will be to build. The project is technically feasible as it uses widely available and well-documented technologies: PHP for server-side scripting, MySQL for data management, and Apache as a web server. The assessment is based on outline design of system requirements in terms of input, processes, output, fields, programs, and procedures. The entire

stack can run on a standard XAMPP environment on any Windows or Linux computer. No specialised hardware is required.

1.3.3 Economical Feasibility

Economical feasibility establishes the cost-effectiveness of the proposed system. All technologies used in this project are open-source and free of cost. The system eliminates the need for physical ticketing infrastructure, reducing operational costs significantly. The purpose of the economic feasibility assessment is to determine the positive economic benefits to the organization that the proposed system will provide. It includes quantification and identification of all the benefits expected. This assessment typically involves a cost/benefits analysis, and the benefits of this project make it economically feasible.

CHAPTER 2

REQUIREMENT ANALYSIS

Requirement analysis is the process of identifying and documenting the needs and expectations of stakeholders for a proposed system. For the Air Ticket Reservation System, the following requirements were identified:

2.1 HARDWARE REQUIREMENTS

Hardware	Specification
Processor	Intel Core i3 or above
Clock Speed	2.0 GHz or higher
RAM Size	4 GB minimum
Hard Disk	100 GB free space
Monitor	15-inch color display
Input Devices	Keyboard and Mouse
Network	Broadband / LAN connection

2.2 SOFTWARE REQUIREMENTS

Software	Specification
Operating System	Windows 10 / Windows 11 / Ubuntu Linux

Web Server	Apache (via XAMPP 8.x)
Browser	Google Chrome (latest version)
Server-side Language	PHP 8.x
Database	MySQL 8.x
Frontend Technologies	HTML5, CSS3, JavaScript (Vanilla)
Development Tool	Visual Studio Code / Notepad++

2.3 FUNCTIONAL REQUIREMENTS

Sr. No.	Module	Functional Requirement
1	User	Users can register with name, email, and password
2	User	Users can login and logout securely using session
3	User	Users can reset password using OTP verification
4	User	Users can search flights by source, destination, and date
5	User	Users can select seats and enter passenger details
6	User	Users can view payment summary and confirm booking
7	User	Users can view their complete booking history
8	Admin	Admin can login and logout securely

9	Admin	Admin can add new flights with all details
10	Admin	Admin can view all flights with pagination
11	Admin	Admin can edit flight fare, seats, status, and delay info
12	Admin	Admin can view all passenger bookings

2.4 NON-FUNCTIONAL REQUIREMENTS

- **Performance:** The system should load each page within 3 seconds on a standard broadband connection.
- **Security:** All pages should be protected by session-based authentication to prevent unauthorised access.
- **Reliability:** The system should handle concurrent users without data inconsistency.
- **Usability:** The interface should be intuitive and easy to navigate for users with basic computer skills.
- **Scalability:** The database should support a large number of flights and booking records with pagination.
- **Maintainability:** Code should be modular and well-organised for easy updates and bug fixing.

CHAPTER 3

PROPOSED SYSTEM

The Air Ticket Reservation System is an online web application that provides the full functionality required to book air tickets. The system allows only registered users to login; new users can register by providing their name, email, and password. This is a web application that runs on a local or hosted Apache server with a MySQL database backend.

Once logged in, a user can search for flights by entering the source city, destination city, and travel date. Available flights are displayed in a table, from which the user selects a flight, chooses the number of seats, fills in passenger details, reviews a payment summary, and confirms the booking. The system automatically deducts booked seats from the available seat count.

The administrator can log in separately to add new flights, edit existing flights (update fare, seats, status, delay information), and view all passenger bookings in a paginated table. The system supports flight delay management where the admin enters delay minutes and the system automatically recalculates the revised departure and arrival times.

3.1 ADVANTAGES OF PROPOSED SYSTEM

- The system allows passengers to book tickets anytime, from anywhere, without visiting a physical office.
- Real-time seat inventory management prevents overbooking and ensures data accuracy.
- Administrators can update flight status (Running / Delayed / Cancelled) instantly.
- Automatic recalculation of delayed departure and arrival times reduces manual effort.
- A delay exceeding 300 minutes triggers automatic cancellation, protecting passenger interests.
- Passengers can view their complete booking history linked to their login email.
- The OTP-based password recovery feature ensures account security.
- City autocomplete using JavaScript improves the search experience for users.

- Paginated views for flights and bookings ensure fast loading even with large datasets.

3.2 COMPARISON: EXISTING VS PROPOSED SYSTEM

Feature	Existing System	Proposed System
Booking Method	Manual / Counter-based	Online via web browser
Data Storage	Paper records / Excel	MySQL relational database
Flight Status Update	Manual announcement	Real-time admin update
Seat Availability	Manually tracked	Auto-decremented on booking
Booking History	Not available	User can view online anytime
Password Recovery	Manual via counter	OTP-based self-service
Access	Office hours only	24/7 online access

CHAPTER 4

DESIGN AND ARCHITECTURE

Design is the first step in the development phase for any techniques and principles for the purpose of defining a device, a process, or system in sufficient detail to permit its physical realisation. Once the software requirements have been analysed and specified, the software design involves three technical activities — Design, Coding, Implementation, and Testing — that are required to build and verify the software. The design activities are of main importance in this phase, because in these activities, decisions ultimately affecting the success of the software implementation and its ease of maintenance are made.

4.1 SYSTEM ARCHITECTURE

The system follows a three-layer client-server architecture:

- **Presentation Layer:** HTML5, CSS3, JavaScript — handles the user interface displayed in the browser. Separate CSS files (user.css and admin.css) provide themed interfaces for users and admins respectively.
- **Application Layer:** PHP scripts — handle all business logic including session management, form processing, database queries, delay calculations, and redirect logic.
- **Data Layer:** MySQL database (air_ticket) — stores all users, flights, and booking records persistently with relational integrity.

4.2 MODULE DESCRIPTION

4.2.1 User Module

The User module covers the complete end-to-end booking journey for passengers. The user can register, login, search for available flights by source/destination/date, select seats, enter passenger details, make payment, and view booking history. The session stores the user ID and email upon successful login, and all subsequent pages are protected by a session check.

4.2.2 Administrator Module

The Admin module is accessible only after successful session login. The administrator can add new flights, view the complete flight list with pagination, edit individual flight details (fare,

seats, status, delay information), and view all passenger bookings across the entire system. The admin dashboard provides a central navigation hub for all these functions.

4.3 ER DIAGRAM

The Entity-Relationship (ER) diagram below shows the three main entities of the system — USERS, FLIGHTS, and BOOKINGS — along with their attributes and relationships. A user **MAKES** bookings, and each booking **BOOKS** a flight. The relationship is one-to-many: one user can make many bookings, and one flight can have many bookings.

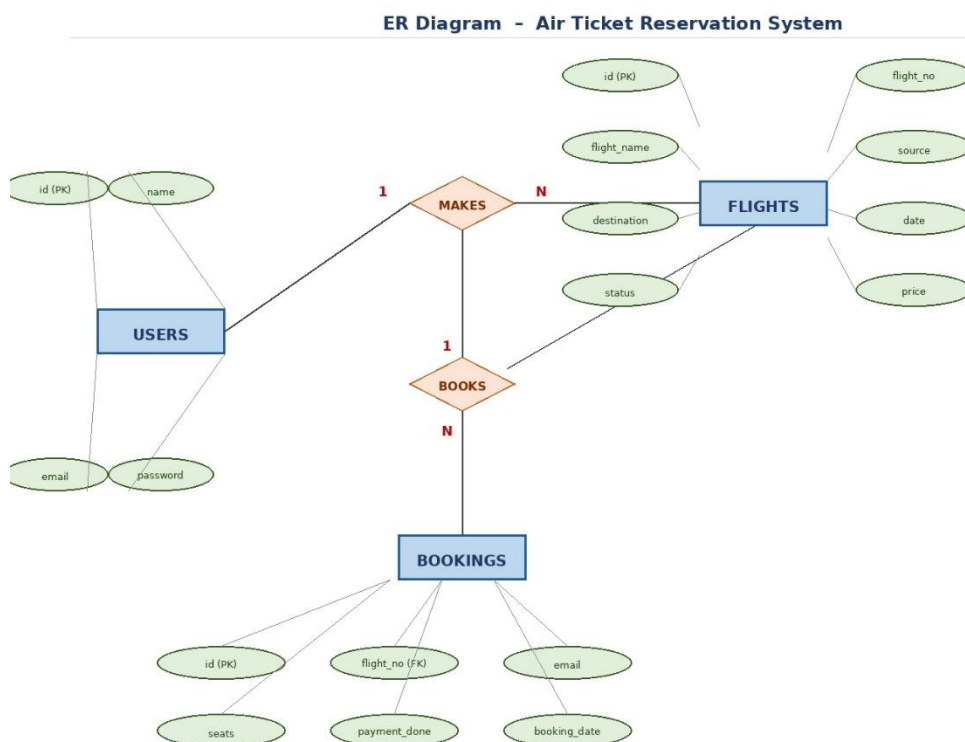


Fig. 4.1: ER Diagram – Air Ticket Reservation System

4.4 DATABASE DESIGN

Database Name: air_ticket. The system uses three primary tables to store all data:

Table 1: users

Sr. No.	Column Name	Data Type	Description
1	id (PK)	INT AUTO_INCREMENT	Unique user identifier
2	name	VARCHAR(100)	Full name of the registered user
3	email	VARCHAR(100)	User login email address
4	password	VARCHAR(100)	User account password

Table 2: flights

Sr. No.	Column Name	Data Type	Description
1	id (PK)	INT	Unique flight record ID
2	flight_no	VARCHAR(20)	Unique flight number
3	flight_name	VARCHAR(100)	Airline or flight name
4	source	VARCHAR(100)	Origin city
5	destination	VARCHAR(100)	Destination city
6	flight_date	DATE	Scheduled date of flight
7	departure_time	TIME	Scheduled departure time
8	arrival_time	TIME	Scheduled arrival time
9	seats	INT	Remaining available seats

10	price	DECIMAL(10,2)	Fare per passenger (INR)
11	status	VARCHAR(20)	Running / Delay / Cancelled
12	delay_minutes	INT	Minutes of delay (if any)
13	d_dep_time	TIME	Revised departure time after delay
14	d_arr_time	TIME	Revised arrival time after delay
15	delay_reason	VARCHAR(255)	Reason for delay

Table 3: bookings

Sr. No.	Column Name	Data Type	Description
1	id (PK)	INT AUTO_INCREMENT	Unique booking ID
2	flight_no	VARCHAR(20)	FK – references flights.flight_no
3	email	VARCHAR(100)	Passenger contact email
4	passenger_names	TEXT	Comma-separated passenger names
5	passenger_ages	TEXT	Comma-separated passenger ages
6	seats	INT	Number of seats booked
7	mobile	VARCHAR(15)	Contact mobile number
8	payment_done	DECIMAL(10,2)	Total amount paid (INR)
9	booking_date	DATETIME	Timestamp of booking

4.5 DFD DIAGRAM

A Data Flow Diagram (DFD) shows how data flows through the system. Level 0 shows the system as a single process with external entities. Level 1 breaks the system into individual processes.

DFD Level 0

The DFD Level 0 (Context Diagram) shows the Air Ticket Reservation System as a single process interacting with two external entities: Admin and User. The Admin sends flight data and receives booking information. The User sends search and booking requests and receives flight results and booking confirmations. The system communicates with the Database for storing and retrieving all data.

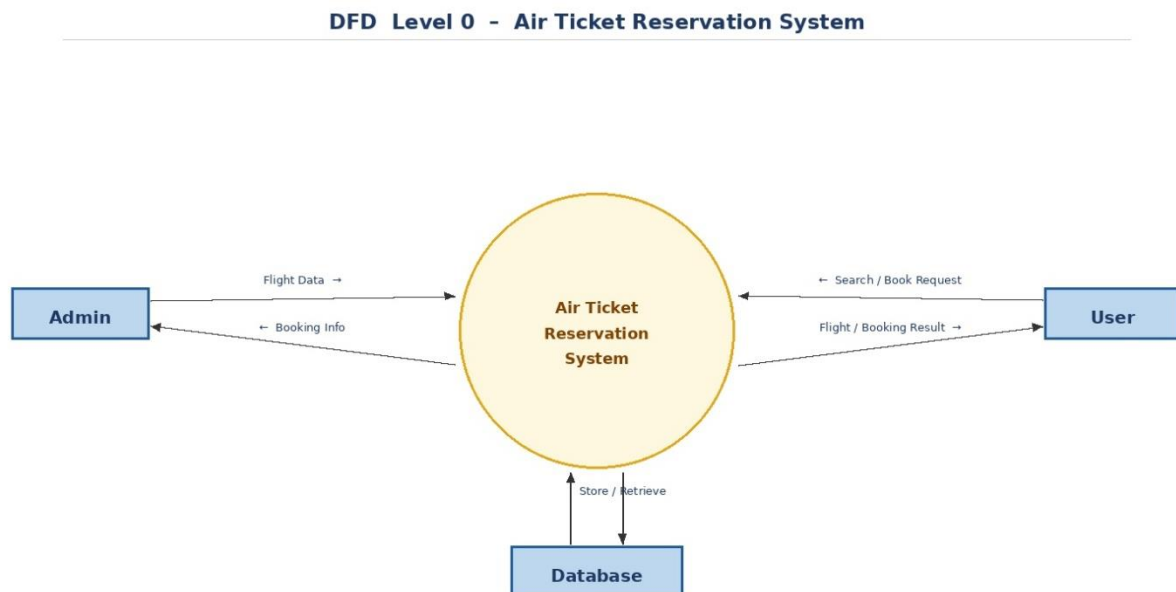


Fig. 4.2: DFD Level 0 – Context Diagram

DFD Level 1

The DFD Level 1 expands the central process into individual sub-processes: Register (1.0), Login (2.0), Search Flights (3.0), Book Ticket (4.0), Payment (5.0), Booking History (6.0),

Manage Flights (7.0), and View Bookings (8.0). Each process interacts with the appropriate data stores (D1: users, D2: flights, D3: bookings) and the respective external entities (User and Admin).

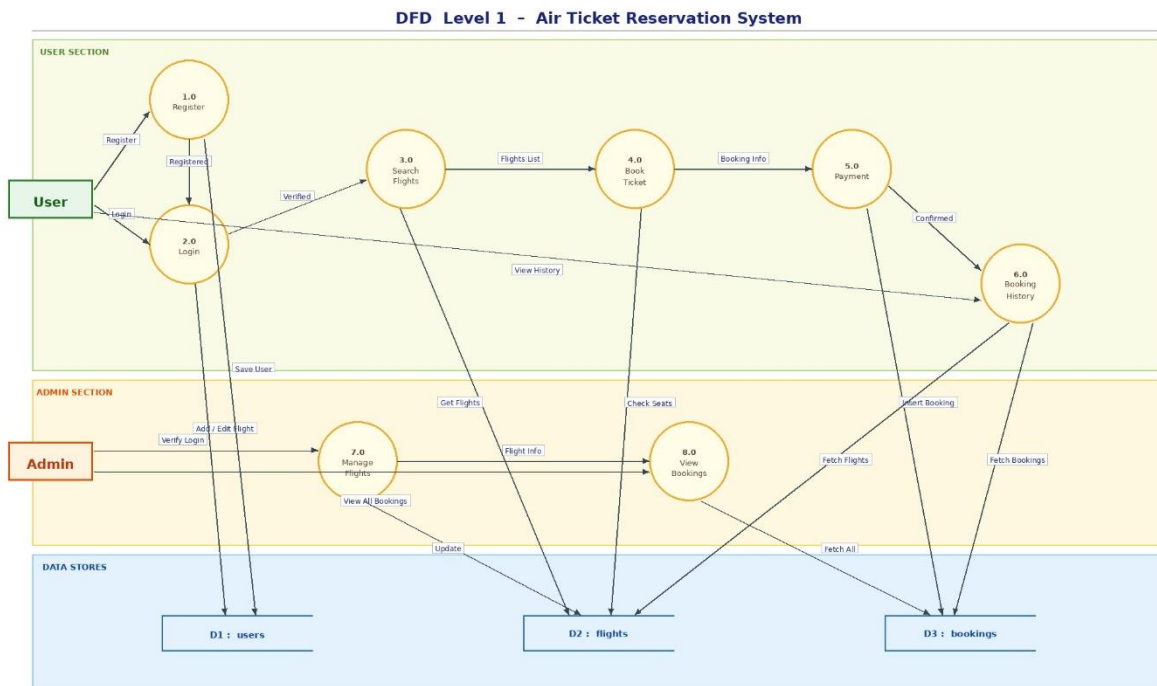


Fig. 4.3: DFD Level 1

4.6 SCREENSHOTS

This section presents the screenshots of all major pages and functionalities of the Air Ticket Reservation System, demonstrating the actual working of the application.

4.6.1 Home Page

The home page (index.php) displays the system title with a runway background image and provides two login buttons — Admin Login and User Login — to direct the respective users to their login pages.

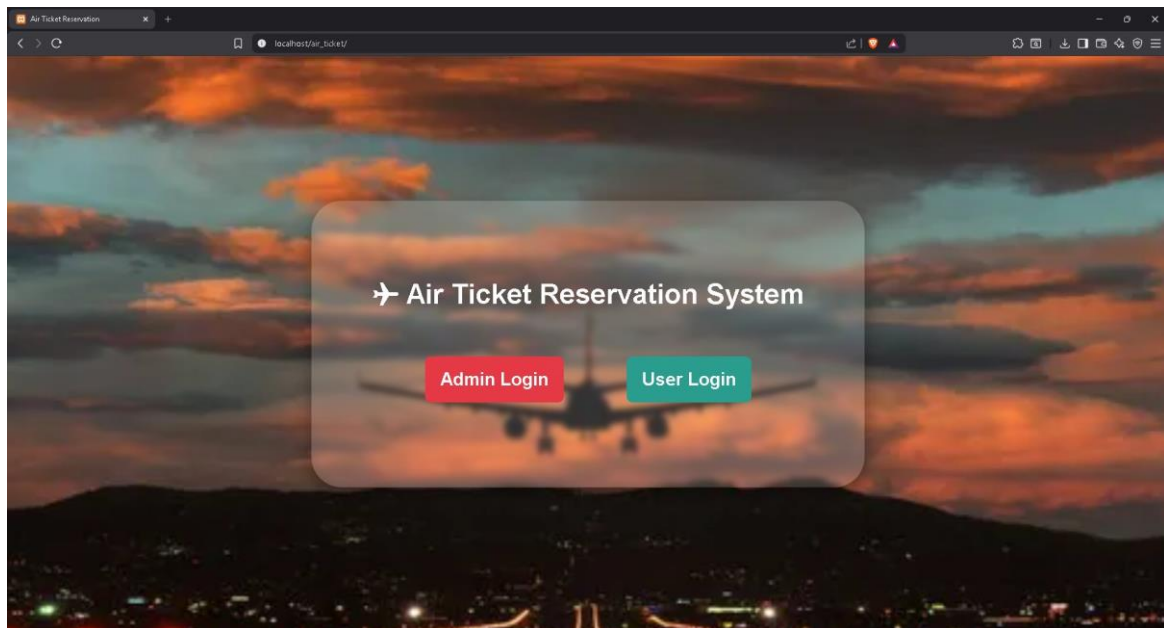


Fig. 4.4.1: Home Page – Air Ticket Reservation System

4.6.2 Admin Login Page

The Admin Login page (admin/login.php) provides a secure login form where the administrator enters their username and password to access the admin panel.

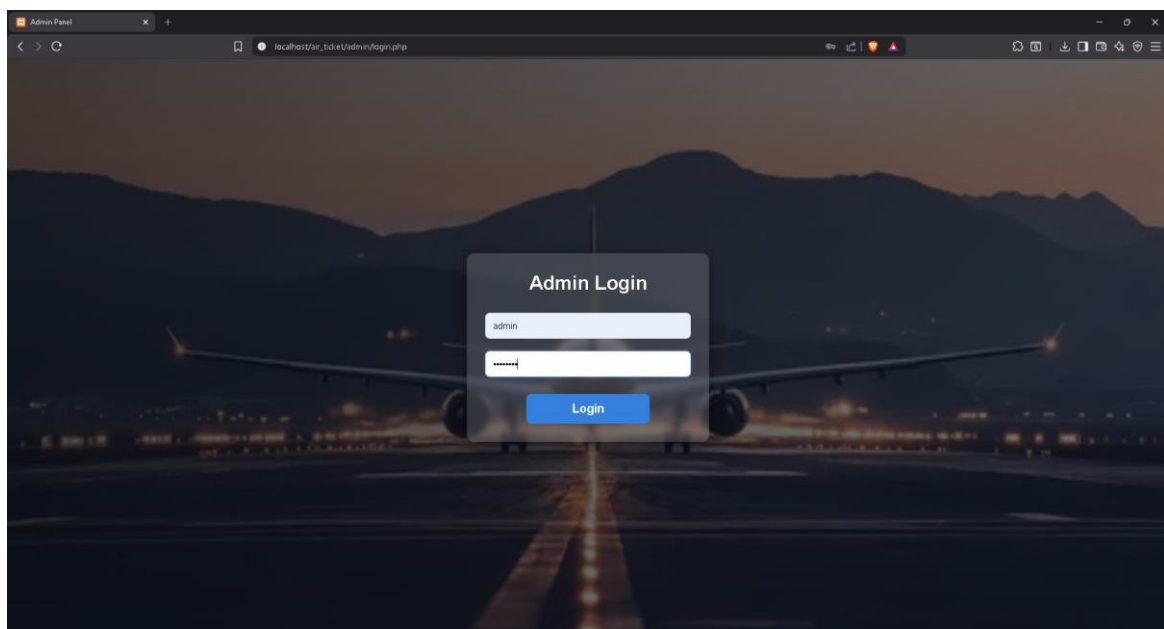


Fig. 4.4.2: Admin Login Page

4.6.3 Admin Dashboard

The Admin Dashboard (admin/dashboard.php) is the central control panel for the administrator. It provides quick navigation to Add Flight, View Flights, View Bookings, and Logout options.

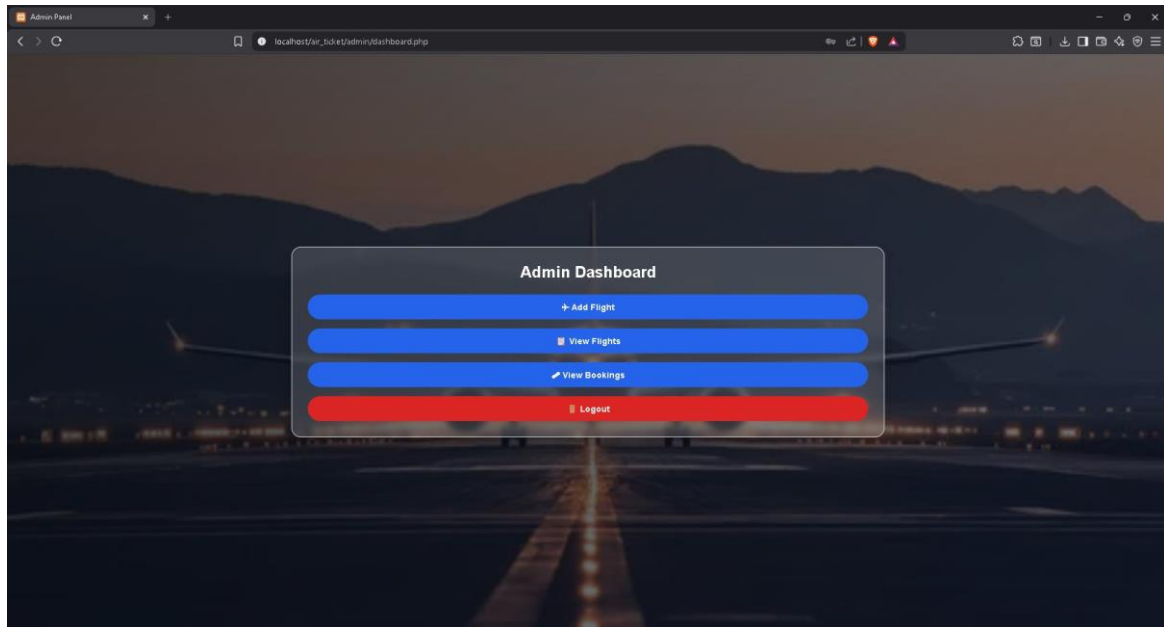


Fig. 4.4.3: Admin Dashboard

4.6.4 Add Flight Form

The Add Flight page (admin/add_flight.php) allows the admin to add a new flight by entering all required details including flight number, name, route, date, departure and arrival time, available seats, and price per passenger.

Fig. 4.4.4: Add Flight Form (Admin)

4.6.5 View Flights (Admin)

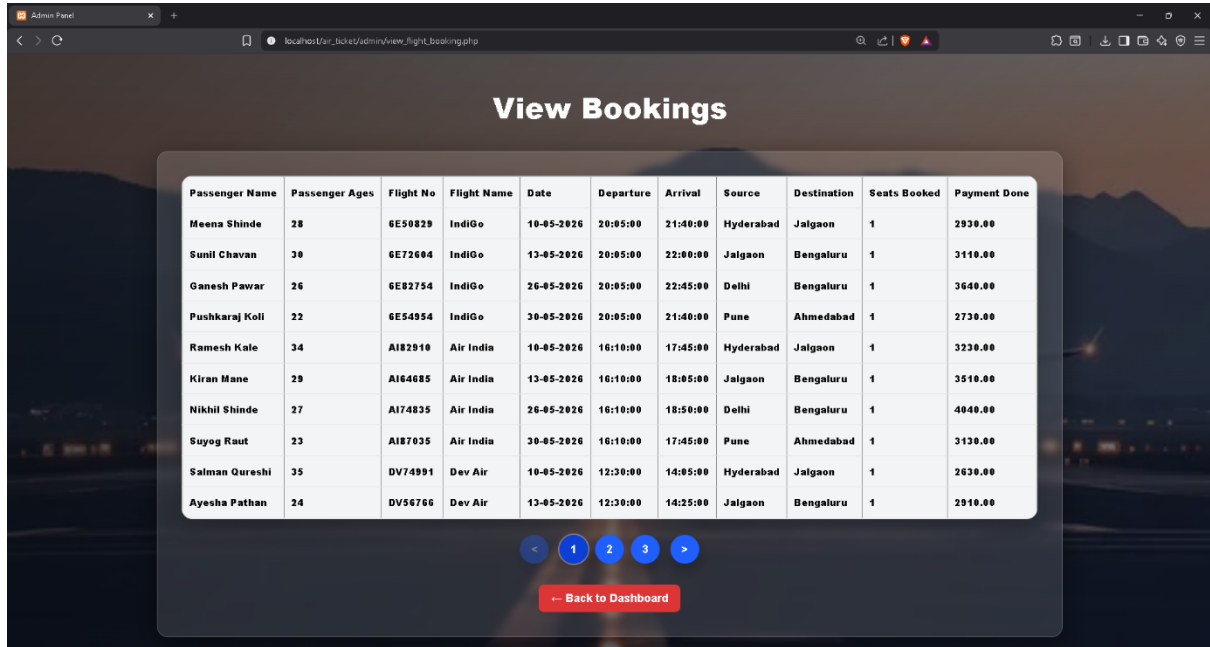
The View Flights page (admin/view_flight.php) displays all flights in a paginated table. Each row shows flight details and an Edit link for updating the flight.

Flight No	Flight Name	Source	Destination	Date	Dep. time	Arr. time	Fare	Status	Action
A178498	Air India	Pune	Goa	2026-04-10	06:30:00	07:40:00	2680.00	Running	Edit
A159153	Air India	Mumbai	Jalgaon	2026-04-10	06:30:00	07:40:00	2780.00	Running	Edit
A176093	Air India	Delhi	Goa	2026-04-10	06:30:00	09:05:00	3870.00	Running	Edit
A158748	Air India	Pune	Jalgaon	2026-04-10	06:30:00	07:45:00	2850.00	Running	Edit
A177688	Air India	Hyderabad	Goa	2026-04-10	06:30:00	08:05:00	3030.00	Running	Edit
A158343	Air India	Delhi	Jalgaon	2026-04-10	06:30:00	08:30:00	3480.00	Running	Edit
A177283	Air India	Bengaluru	Goa	2026-04-10	06:30:00	07:55:00	2890.00	Running	Edit
A157938	Air India	Hyderabad	Jalgaon	2026-04-10	06:30:00	08:05:00	3130.00	Running	Edit
A176878	Air India	Ahmedabad	Goa	2026-04-10	06:30:00	08:05:00	3030.00	Running	Edit
A157533	Air India	Bengaluru	Jalgaon	2026-04-10	06:30:00	08:25:00	3410.00	Running	Edit
A157128	Air India	Ahmedabad	Jalgaon	2026-04-10	06:30:00	07:50:00	2920.00	Running	Edit
A156723	Air India	Goa	Jalgaon	2026-04-10	06:30:00	07:45:00	2850.00	Running	Edit
A156318	Air India	Jalgaon	Mumbai	2026-04-10	06:30:00	07:40:00	2780.00	Running	Edit
A155913	Air India	Pune	Mumbai	2026-04-10	06:30:00	07:30:00	2540.00	Running	Edit
A155508	Air India	Delhi	Mumbai	2026-04-10	06:30:00	08:35:00	3450.00	Running	Edit

Fig. 4.4.5: View Flights (Admin Panel)

4.6.6 Admin View Bookings

The Admin View Bookings page provides the administrator with a paginated view of all passenger bookings across the entire system.



View Bookings

Passenger Name	Passenger Ages	Flight No	Flight Name	Date	Departure	Arrival	Source	Destination	Seats Booked	Payment Done
Meena Shinde	28	6E50829	IndiGo	10-05-2026	20:05:00	21:40:00	Hyderabad	Jalgaon	1	2930.00
Sunil Chavan	30	6E72604	IndiGo	13-05-2026	20:05:00	22:00:00	Jalgaon	Bengaluru	1	3110.00
Ganesh Pawar	26	6E82754	IndiGo	26-05-2026	20:05:00	22:45:00	Delhi	Bengaluru	1	3640.00
Pushkaraj Koli	22	6E54954	IndiGo	30-05-2026	20:05:00	21:40:00	Pune	Ahmedabad	1	2730.00
Ramesh Kale	34	AI82910	Air India	10-05-2026	16:10:00	17:45:00	Hyderabad	Jalgaon	1	3230.00
Kiran Mane	29	AI64685	Air India	13-05-2026	16:10:00	18:05:00	Jalgaon	Bengaluru	1	3510.00
Nikhil Shinde	27	AI74835	Air India	26-05-2026	16:10:00	18:50:00	Delhi	Bengaluru	1	4040.00
Suyog Raut	23	AI87035	Air India	30-05-2026	16:10:00	17:45:00	Pune	Ahmedabad	1	3130.00
Salman Qureshi	35	DV74991	Dev Air	10-05-2026	12:30:00	14:05:00	Hyderabad	Jalgaon	1	2630.00
Ayesha Pathan	24	DV56766	Dev Air	13-05-2026	12:30:00	14:25:00	Jalgaon	Bengaluru	1	2910.00

[<](#)
[1](#)
[2](#)
[3](#)
[>](#)

[← Back to Dashboard](#)

Fig. 4.4.6: Admin – View All Bookings

4.6.7 User Registration

The User Registration page (user/register.php) allows new users to create an account by entering their name, email address, and password.

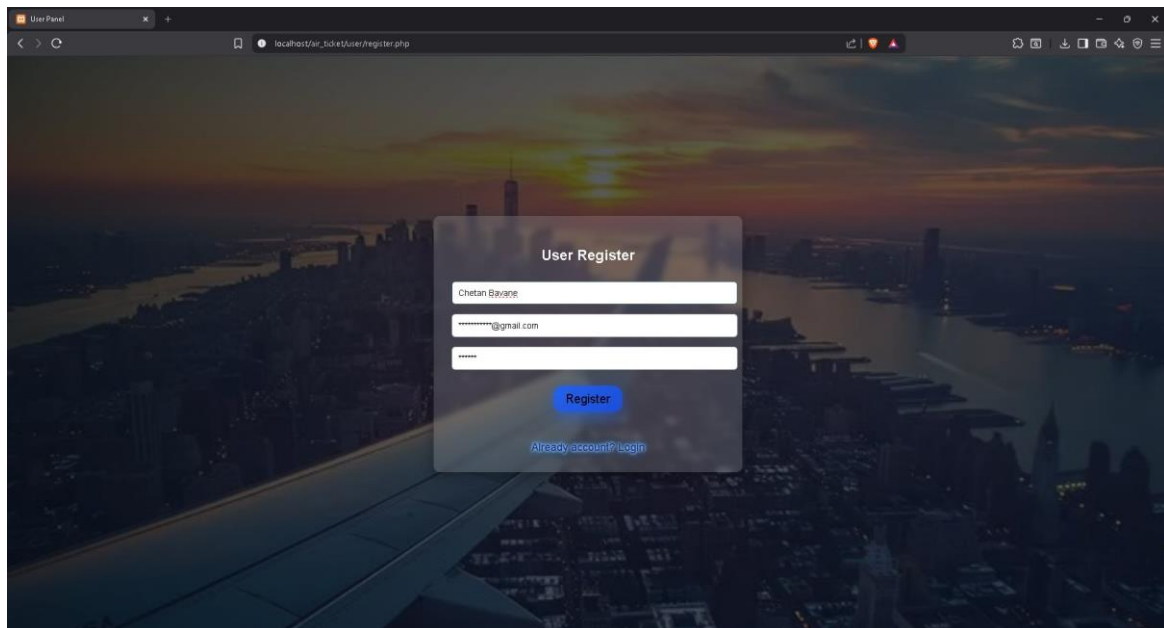


Fig. 4.4.7: User Registration Page

4.6.8 Forgot Password / OTP

The Forgot Password page enables users to reset their password using a 6-digit OTP generated server-side and displayed on-screen for demo purposes.

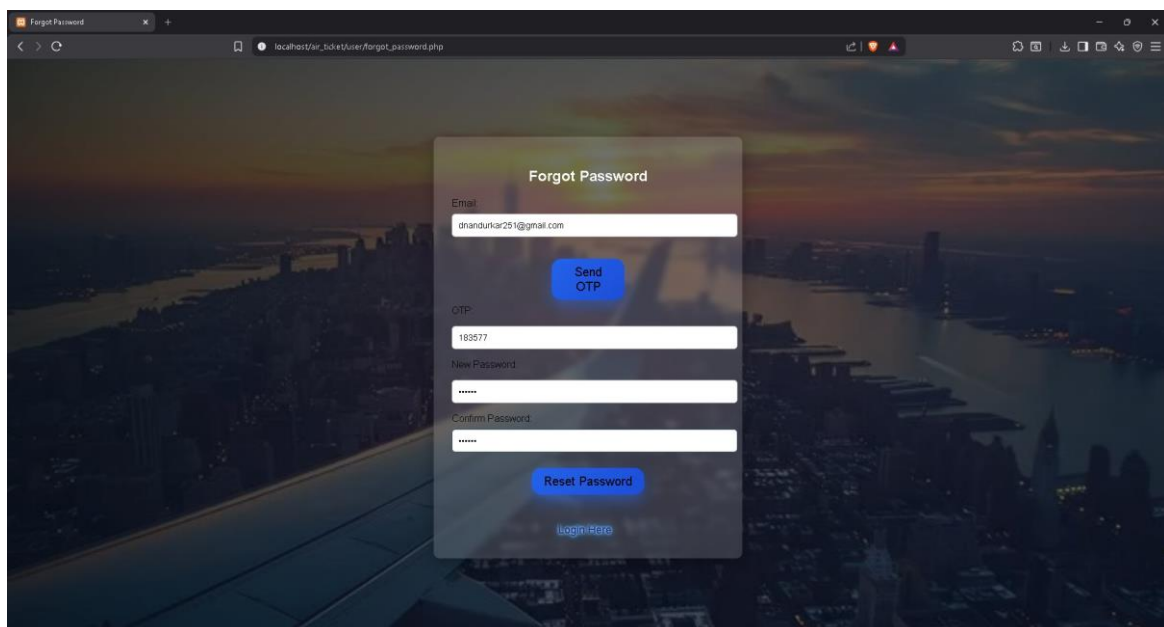


Fig. 4.4.8: Forgot Password / OTP Verification Page

4.6.9 Search Flights

The Search Flights page allows logged-in users to search for flights by entering source city, destination city, and travel date. Results are displayed with a Book button for each available flight.

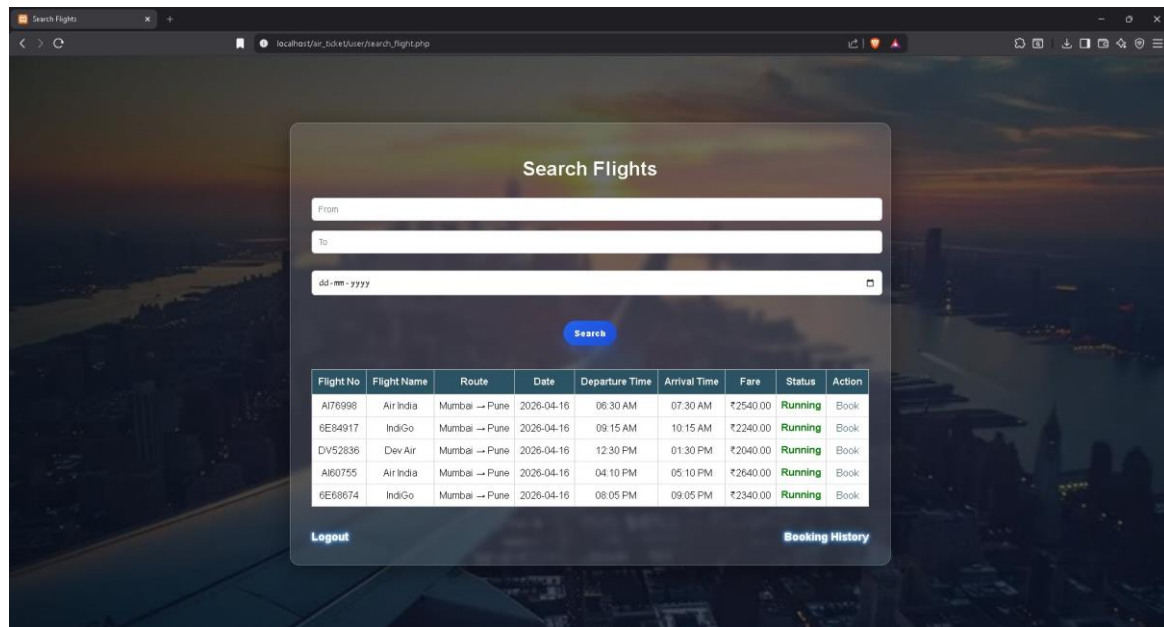


Fig. 4.4.9: Search Flights Page with Results

4.6.10 Passenger Details Form

The Passenger Details page dynamically generates input fields for each passenger based on seats selected. The user also enters contact email and mobile number.

Passenger Details

Flight: DV52836
Flight: Dev Air
Seats Selected: 1

Passenger 1

Name: Devendra Nandurkar
Age: 21

Contact Details

Email: [redacted]@gmail.com
Mobile No: [redacted]

Confirm & Go to Payment

Fig. 4.4.10: Passenger Details Form

4.6.11 Payment Window

The Payment page displays a booking summary showing flight details, passenger names, per-passenger fare, and total amount. The user confirms and completes the booking.

Payment Window

Booking Details

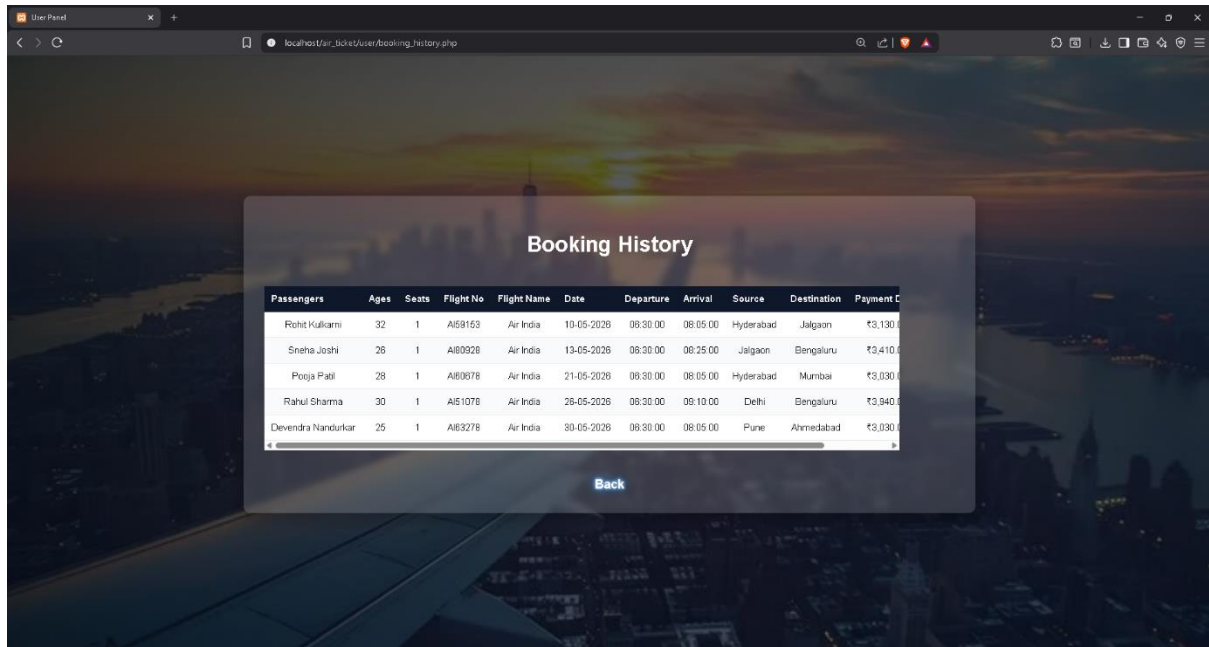
Flight No	Flight Name	Passenger Name	Passenger Age	Price (Per Passenger)
DV52836	Dev Air	Devendra Nandurkar	21	₹2040.00
Total Price				₹2040

Pay & Book Ticket

Fig. 4.4.11: Payment Window

4.6.12 Booking History (User)

The Booking History page shows all bookings made by the logged-in user, including passenger details, flight information, route, amount paid, and current flight status.



Passengers	Ages	Seats	Flight No	Flight Name	Date	Departure	Arrival	Source	Destination	Payment C
Rohit Kulkarni	32	1	AI59153	Air India	10-05-2026	08:30:00	08:05:00	Hyderabad	Jalgaon	₹3,130.00
Sneha Joshi	26	1	AI80928	Air India	13-05-2026	08:30:00	06:25:00	Jalgaon	Bengaluru	₹3,410.00
Pooja Patil	28	1	AI80876	Air India	21-05-2026	08:30:00	08:05:00	Hyderabad	Mumbai	₹3,030.00
Rahul Sharma	30	1	AI51076	Air India	26-05-2026	08:30:00	08:10:00	Delhi	Bengaluru	₹3,940.00
Devendra Nandurkar	25	1	AI83278	Air India	30-05-2026	08:30:00	08:05:00	Pune	Ahmedabad	₹3,030.00

Back

Fig. 4.4.12: Booking History – Single Booking

CHAPTER 5

DEVELOPMENT AND CODING

The development phase involves writing the actual code for the system based on the design specifications. The Air Ticket Reservation System was developed using PHP for server-side logic, MySQL for database operations, HTML and CSS for the user interface, and JavaScript for client-side interactivity.

5.1 TECHNOLOGY DESCRIPTION

5.1.1 PHP

PHP is a server-side scripting language designed primarily for web development. Originally created by Rasmus Lerdorf in 1994, PHP stands for the recursive acronym PHP: Hypertext Preprocessor. PHP code may be embedded into HTML or HTML5 code, or it can be used in combination with various web template systems and web frameworks. PHP code is usually processed by a PHP interpreter implemented as a module in the web server. In this project, PHP 8.x is used to handle all business logic including session management, database interaction, form processing, and redirect logic.

PHP was chosen for this project because it is open-source, runs on all major web servers (Apache, Nginx, IIS), and has excellent MySQL integration through the MySQLi extension. PHP's session handling functions (`$_SESSION`) are used to maintain login state across pages for both admin and user roles. The `DateTime` class in PHP is used for delay time recalculation in the flight management module.

5.1.2 HTML and CSS

HyperText Markup Language (HTML) is the standard markup language for creating web pages. HTML elements are the building blocks of HTML pages. In this project, HTML5 is used to structure all web pages including forms for login, registration, flight search, seat selection, and passenger details. CSS3 (Cascading Style Sheets) is used to style all pages. The project uses two separate CSS files — `admin.css` for the admin panel and `user.css` for the user interface — each providing a distinct themed design with background images, card layouts, and styled buttons.

5.1.3 MySQL

MySQL is an open-source relational database management system (RDBMS). MySQL is a central component of the LAMP open-source web application software stack. In this project, MySQL 8.x is used to manage the air_ticket database containing three tables: users, flights, and bookings. MySQLi (MySQL Improved) extension in PHP is used for all database queries. The booking_history.php page uses prepared statements for secure parameterised queries.

5.1.4 JavaScript

JavaScript is used in this project for client-side interactivity. The primary JavaScript feature is the city autocomplete function on the search_flight.php page. A predefined array of city names (Jalgaon, Mumbai, Pune, Delhi, Hyderabad, Bengaluru, Ahmedabad, Goa) is filtered on keypress to show matching suggestions in a dropdown. JavaScript is also used for showing alert dialogs for booking confirmation and error messages via PHP-generated script tags.

5.2 ADMIN MODULE

The Admin module is accessible only after a successful session login via admin/login.php. The admin credentials are verified against a hardcoded check. Once authenticated, the administrator is redirected to the dashboard. The Admin Dashboard provides three core navigation options: Add Flight, View Flights, and View Bookings. Each option is linked to its respective PHP page. A Logout button destroys the admin session.

5.2.1 Add Flight

The add_flight.php page allows the admin to enter all flight details. Before inserting, the system checks whether the flight number already exists. If it does, an alert is shown and the insertion is blocked. The departure and arrival times are assembled from separate hour, minute, and AM/PM dropdown selections into a single time string before being stored.

5.2.2 View Flights

The view_flight.php page displays all flights in a paginated table with 15 rows per page sorted by date and departure time. The system checks whether any flight has delay data; if so, two additional columns (Delayed Departure and Delayed Arrival) are shown automatically using a conditional query.

5.2.3 Edit Flight

The `edit_flight.php` page allows the admin to update fare, available seats, status, delay minutes, and delay reason. When the status is set to Delay, PHP's `DateTime` class adds the delay minutes to the original times to produce revised departure and arrival times. If the delay exceeds 300 minutes, the status is automatically changed to Cancelled and delay fields are cleared.

5.3 USER MODULE

5.3.1 User Registration and Login

New users register via `user/register.php` by entering their name, email, and password. On login, the system queries the users table to verify credentials. On success, the user ID and email are stored in the PHP session and the user is redirected to the search page.

5.3.2 Forgot Password

The `forgot_password.php` page generates a random 6-digit OTP using PHP's `rand()` function, stores it in the session, and displays it on screen. After verifying the OTP, the user sets a new password which is updated in the users table.

5.3.3 Search Flights

The `search_flight.php` page runs a MySQL query filtering flights by source, destination, date, and status (excluding Cancelled). JavaScript autocomplete suggests city names as the user types in the From and To fields.

5.3.4 Booking Process

The booking process spans three pages. `book_ticket.php`: user chooses number of seats (limited to available). `passenger_details.php`: user enters name and age for each passenger plus contact details. `payment.php`: shows summary; on Pay & Book Ticket, booking is inserted into the database, seats are decremented, and sessions are cleared.

5.3.5 Booking History

The booking_history.php page uses MySQLi prepared statements to fetch all bookings for the logged-in email, joined with the flights table. Delay columns appear conditionally only when delayed bookings exist.

5.4 KEY FEATURES SUMMARY

Sr. No.	Feature	Details
1	Session Authentication	Separate admin and user sessions; redirect to login if session missing
2	City Autocomplete	JavaScript filters predefined city list on keypress
3	Duplicate Flight Guard	Admin cannot add a flight with an existing flight number
4	Delay Management	Auto-calculates revised departure and arrival times
5	Auto-Cancellation	Delays exceeding 300 minutes trigger automatic cancellation
6	Conditional Columns	Delay columns appear only when delay data exists
7	Seat Inventory	Booking deducts seats; user cannot exceed available seats
8	Pagination	All listing pages use server-side pagination
9	OTP Password Reset	6-digit OTP generated server-side
10	Responsive CSS Themes	Separate admin.css and user.css for distinct interfaces

CHAPTER 6

TESTING AND IMPLEMENTATION

Testing is a critical phase of software development that ensures the system meets its requirements and functions correctly. The term implementation has different meanings ranging from the conversion of a basic application to a complete replacement of a computer system. For the Air Ticket Reservation System, multiple testing strategies were applied to verify correctness across all modules.

6.1 UNIT TESTING

In computer programming, unit testing is a software testing method by which individual units of source code are tested to determine whether they are fit for use. Individual PHP functions and SQL queries were tested in isolation. The flight search query filter (source, destination, date, status exclusion) was tested with various combinations. The booking insert query was verified with test passenger data. The seat update query was confirmed to correctly decrement the seats column after each booking.

- Find problems early: Unit testing finds problems early in the development cycle.
- Facilitates Change: Unit testing allows the programmer to refactor code safely.
- Simplifies Integration: By testing parts first, integration testing becomes much easier.
- Documentation: Unit tests serve as living documentation of the system behaviour.

6.2 INTEGRATION TESTING

Integration testing is the phase in software testing in which individual software modules are combined and tested as a group. It occurs after unit testing. Integration testing ensured that data flowed correctly across all pages in the booking workflow: from flight search to seat selection, passenger details, payment confirmation, and booking history. Session variables (`$_SESSION['seat_count']` and `$_SESSION['payment']`) were inspected at each step to confirm correct data passing between pages.

6.3 BLACK-BOX TESTING

Black-box testing is a method of software testing that examines the functionality of an application without peering into its internal structures or workings. Test cases were designed for all user-facing features including registration, login, flight search, booking, OTP reset, and

admin operations. The tester is aware of what the software is supposed to do but is not aware of how it does it. Typical black-box test design techniques include: decision table testing, equivalence partitioning, boundary value analysis, cause-effect graph, state transition testing, and use case testing.

6.4 WHITE-BOX TESTING

White-box testing is a method of testing software that tests internal structures or workings of an application. It was applied to verify the delay time recalculation algorithm using PHP's DateTime class, the auto-cancellation trigger when delay exceeds 300 minutes, seat inventory decrement on booking confirmation, and session variable management across page redirections. White-box test design techniques include control flow testing, data flow testing, branch testing, statement coverage, and decision coverage.

- Side effects of having the knowledge of the source code is beneficial to thorough testing.
- Optimization of code by revealing hidden errors and being able to remove possible defects.
- Provides traceability of tests from the source, allowing future changes to be easily captured.
- White-box tests are easy to automate.

6.5 SYSTEM TESTING

System testing of software is testing conducted on a complete, integrated system to evaluate the system's compliance with its specified requirements. System testing falls within the scope of black-box testing. The complete application was tested end-to-end on a local XAMPP server. Both the admin and user workflows were tested for correctness, navigation, and edge cases such as booking with maximum seats, attempting to book a cancelled flight, and verifying incorrect OTP rejection.

6.6 TEST CASES

Sr. No.	Test Case	Input	Expected Result	Status

1	User Registration	Valid name, email, password	Account created; redirect to login	Pass
2	User Login – Valid	Correct email and password	Session created; redirect to search	Pass
3	User Login – Invalid	Wrong password	Alert: Invalid Email or Password	Pass
4	Search Available Flight	Valid source, dest, date	Matching non-cancelled flights shown	Pass
5	Search No Results	No flights for that route	Message: No flights available	Pass
6	Exceed Available Seats	seats > available	Error: Seats not available	Pass
7	Complete Booking	All passenger details	Booking saved; seats decremented	Pass
8	Duplicate Flight No.	Existing flight_no	Alert: Flight ID already exists	Pass
9	Delay > 300 Minutes	delay_minutes = 350	Flight auto-cancelled	Pass
10	Valid OTP Entry	Correct 6-digit OTP	Password updated successfully	Pass
11	Invalid OTP Entry	Wrong OTP	Alert: Invalid OTP	Pass
12	View Booking History	Logged-in user email	All bookings for that email shown	Pass

CHAPTER 7

CONCLUSION

The Air Ticket Reservation System is a fully functional web-based application that covers the complete lifecycle of airline ticket booking. Our project is a humble venture to satisfy the needs of managing flight reservations digitally. Several user-friendly coding approaches have also been adopted. The objective of software planning is to provide a framework that enables the manager to make reasonable estimates at the beginning of the software project.

The system was developed with a clean separation of admin and user roles, each protected by session-based authentication. The admin panel provides complete control over flight management including real-time delay and cancellation handling with automatic time recalculation. The user interface is intuitive and provides a seamless booking experience from login to payment confirmation.

At the end of this project, the following objectives were successfully achieved:

- A description of background and context of the project and its relation to work already done in the area.
- A user-friendly interface was developed for both passengers and administrators with themed CSS designs.
- A secure session-based login system was implemented for two separate roles.
- Real-time flight search, seat selection, and booking were implemented end-to-end.
- Flight delay and cancellation management with automatic time recalculation was built.
- Passenger booking history linked to login email was implemented with pagination.
- OTP-based password recovery was implemented for user account security.
- The system was designed user interface and security issues related to the system.
- Finally, the system is implemented and tested according to the test cases.

CHAPTER 8

FUTURE WORK

It can be summarised that the future scope of the project circles around maintaining information and adding the following enhancements to increase applicability and usage of the project:

- Password Security: Implement password hashing using PHP's `password_hash()` and `password_verify()` functions to protect user passwords from being stored as plain text in the database.
- SQL Injection Prevention: Migrate all SQL queries throughout the application to MySQLi prepared statements to prevent SQL injection vulnerabilities.
- Real Email OTP: Integrate real email OTP delivery using PHPMailer or an SMTP service instead of displaying the OTP on screen.
- Payment Gateway: Connect a real payment gateway such as Razorpay or PayU for actual financial transactions.
- E-Ticket Generation: Add automatic e-ticket PDF generation and email delivery after booking confirmation.
- Seat Map Selection: Introduce a visual seat-map selection feature for passengers to choose specific seat numbers on the aircraft.
- We will host the platform on online servers to make it accessible worldwide.
- Integrate multiple load balancers to distribute the loads of the system.
- Create the master and slave database structure to reduce the overload of the database queries.
- Implementing the backup mechanism for taking backup on codebase and database on regular basis on different servers.
- Build a REST API layer to support a mobile application (Android/iOS) for the same system.
- Add an admin analytics dashboard showing total bookings per day, revenue, most popular routes, and seat occupancy rates using charts.

CHAPTER 9

REFERENCES

9.1 Books

1. PHP & MySQL: Novice to Ninja – Kevin Yank, SitePoint Publications
2. Learning PHP, MySQL & JavaScript – Robin Nixon, O'Reilly Media
3. PHP and MySQL Web Development – Luke Welling & Laura Thomson, Addison-Wesley
4. HTML and CSS: Design and Build Websites – Jon Duckett, Wiley

9.2 Online References

1. <http://www.w3schools.com/>
2. <http://stackoverflow.com/>
3. <https://www.tutorialspoint.com/php/>
4. <http://github.com/>
5. <https://dev.mysql.com/doc/>
6. <https://www.php.net/manual/en/>